GRADE 2 · EVENTS & PARALLEL SCRIPTS

When I Receive

The message sets the order

Name: .... Date: ....

★ **I can trace which script runs first and which runs when it receives a message.**

Who goes first?

Bit starts on the green flag and ends by broadcasting 'start'. Pixel has the hat 'when I receive start', so Pixel waits until that message comes. That sets the ORDER: Bit runs first, then Pixel runs when it hears the message. Read both scripts and find the order.

The number path

0	1	2	3	4	5
Pixel	Bit				

Bit

Bit's script

⚑ when green flag clicked

forward 3

broadcast 'start'

Pixel

Pixel's script

when I receive 'start'

forward 2

1 **Which script runs first?**

Which sprite's script runs FIRST — Bit or Pixel? Write it, and write what event starts it.

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2 **Which runs second?**

Which sprite runs SECOND, and what makes it start? Write the sprite and the event.

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3 **Where do they land?**

Bit runs from home 1: forward 3. Pixel runs from home 0: forward 2. Where does each land? Bit: ____ Pixel: ____

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Big idea

A 'when I receive' script does not start until its message arrives. So a broadcast can set the ORDER of two scripts: the sender runs first, and the receiver runs when the message reaches it.

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When I Receive

Quick facilitation notes & answer key

What this worksheet teaches

Students trace the ORDER that event-started moves run in. Bit runs on the green flag and sends a message; Pixel waits for that message, so it runs second. The message decides who goes first and who goes next. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel), and a card that says "go" you can hand over to act out sending a message.

How to lead it (about 20 minutes)

1. Show them (you do). Bit runs first (green flag), sends "start"; Pixel waits, so it runs second.
2. Do one together. Exercise 1 — Bit runs first; Exercise 2 — Pixel runs second when it gets the message.
3. Let them try. Students find where each lands (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit runs first; its event is "when green flag clicked".

Exercise 2 — Pixel runs second; it starts "when I receive start" (the message Bit sent).

Exercise 3 — Bit: 1, forward 3 → 4. Pixel: 0, forward 2 → 2. Even though there is one green flag, Pixel waits for the message, so the order is Bit then Pixel.

If students get stuck — and ways to adjust

Watch for: thinking both start together. Pixel waits for the message, so it is second.

Make it easier: act it out — Bit moves, hands over the "start" card, then Pixel moves.

Make it harder: ask what would make Pixel run first instead.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions that are started by an event, like clicking the green flag or getting a message.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change what starts them or one move, and explain what the change did.

You'll know it worked when

The child names the run order (Bit then Pixel) and finds both landings.